Practical Assignment: Sales Dashboard Using Power BI

1. Title

Design and Develop an Interactive Sales Dashboard using Power BI

2. Aim

To design an interactive **Sales Dashboard using Microsoft Power BI** for analyzing sales, profit, quantity, regional performance, product performance, and monthly sales trends.

3. Learning Objectives

After completing this practical, students will be able to:

1. Import data into Power BI.
2. Clean and transform data using Power Query.
3. Create calculated columns and DAX measures.
4. Create different types of data visualizations.
5. Use slicers and filters for interactive analysis.
6. Design an interactive dashboard.
7. Interpret business insights from sales data.

4. Problem Statement

A retail company sells different products across different regions of India. The company wants to analyze its sales performance.

Management wants to know:

- What is the **total sales**?
- What is the **total profit**?
- Which **category** generates the highest sales?
- Which **region** performs best?
- Which products are the **top-selling products**?
- How do sales change **month by month**?
- Which salesperson generates the highest sales?
- What is the overall **profit margin**?

Develop an interactive **Power BI Sales Dashboard** that answers these questions.

5. Sample Dataset

Students can create an Excel file named:

Sales_Data.xlsx

Create a worksheet named **Sales**.

Enter the following data:

Order ID	Order Date	Product	Category	Region	Salesperson	Quantity	Unit Price	Sales	Profit
ORD001	05-01-2026	Laptop	Electronics	West	Amit	2	50000	100000	12000
ORD002	08-01-2026	Mobile	Electronics	South	Priya	5	20000	100000	15000
ORD003	12-01-2026	Chair	Furniture	North	Rahul	10	3000	30000	5000
ORD004	18-01-2026	Table	Furniture	West	Sneha	4	7000	28000	4500
ORD005	25-01-2026	Headphones	Accessories	East	Amit	15	1500	22500	4500
ORD006	03-02-2026	Laptop	Electronics	South	Priya	3	50000	150000	18000
ORD007	10-02-2026	Mobile	Electronics	West	Rahul	8	20000	160000	24000
ORD008	15-02-2026	Chair	Furniture	East	Sneha	12	3000	36000	6000
ORD009	21-02-2026	Table	Furniture	North	Amit	6	7000	42000	7000
ORD010	28-02-2026	Headphones	Accessories	South	Priya	20	1500	30000	6000
ORD011	05-03-2026	Laptop	Electronics	North	Rahul	4	50000	200000	24000
ORD012	09-03-2026	Mobile	Electronics	East	Sneha	7	20000	140000	21000
ORD013	16-03-2026	Chair	Furniture	West	Amit	15	3000	45000	7500
ORD014	22-03-2026	Table	Furniture	South	Priya	8	7000	56000	9000
ORD015	29-03-2026	Headphones	Accessories	North	Rahul	25	1500	37500	7500

Note: For a more meaningful dashboard, students can expand this dataset to 100–500 records.

6. Software Required

- **Microsoft Power BI Desktop**
- Microsoft Excel

Power BI Desktop can be used for creating the report and dashboard.

7. Procedure

Step 1: Prepare the Excel Dataset

Create the Excel file:

Sales_Data.xlsx

Enter the sales data in a worksheet named **Sales**.

Save the file.

Step 2: Open Power BI Desktop

Open **Power BI Desktop**.

Select:

Home → Get Data → Excel Workbook

Select:

`Sales_Data.xlsx`

Select the **Sales** worksheet and click:

Load

The sales data will appear in Power BI.

8. Data Cleaning Using Power Query

Select:

Home → Transform Data

Power Query Editor will open.

Check the following:

Order ID

Data type:

Text

Order Date

Data type:

Date

Product

Data type:

Text

Category

Data type:

Text

Region

Data type:

Text

Quantity

Data type:

Whole Number

Unit Price

Data type:

Decimal Number

Sales

Data type:

Decimal Number

Profit

Data type:

Decimal Number

Check whether there are:

- Missing values
- Duplicate records
- Incorrect data types
- Spelling inconsistencies

After cleaning:

Home → Close & Apply

9. Create DAX Measures

Go to:

Modeling → New Measure

Create the following measures.

9.1 Total Sales

```
Total Sales = SUM(Sales[Sales])
```

This calculates the total sales value.

9.2 Total Profit

```
Total Profit = SUM(Sales[Profit])
```

This calculates the total profit.

9.3 Total Quantity

```
Total Quantity = SUM(Sales[Quantity])
```

This calculates the total number of products sold.

9.4 Total Orders

```
Total Orders = DISTINCTCOUNT(Sales[Order ID])
```

This calculates the number of unique orders.

9.5 Average Sales

```
Average Sales = AVERAGE(Sales[Sales])
```

9.6 Profit Margin

```
Profit Margin =
DIVIDE([Total Profit], [Total Sales], 0)
```

Format this measure as **Percentage**.

10. Create Sales Dashboard

Create a new report page.

Rename the page:

Sales Dashboard

The dashboard can be divided into four sections.

Section A: KPI Cards

Create four **Card** visuals.

Card 1 – Total Sales

Field:

Total Sales

Example:

₹1,187,000

Card 2 – Total Profit

Field:

Total Profit

Example:

₹1,57,000

Card 3 – Total Orders

Field:

Total Orders

Example:

15

Card 4 – Profit Margin

Field:

Profit Margin

Example:

13.23%

The actual values will depend on the dataset.

11. Monthly Sales Trend

Use a **Line Chart**.

X-axis

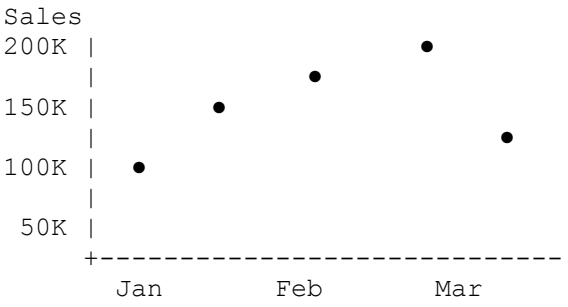
Order Date

Y-axis

Total Sales

The resulting chart will show how sales change over time.

Example:



Interpretation

The line chart helps management identify:

- Increasing sales
- Decreasing sales
- High-sales months
- Low-sales months
- Seasonal patterns

12. Sales by Category

Create a **Clustered Column Chart**.

X-axis

Category

Y-axis

Total Sales

Example:

Category	Sales
Electronics	₹850,000
Furniture	₹200,000
Accessories	₹137,000

The chart allows comparison between product categories.

13. Sales by Region

Create a **Bar Chart**.

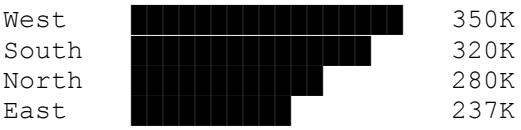
Y-axis

Region

X-axis

Total Sales

Example:



Interpretation

The visualization identifies the best-performing geographical region.

14. Top 5 Products

Create a **Bar Chart**.

Y-axis

Product

X-axis

Total Sales

Use a visual-level filter:

Product → Top N → Top 5 → By Total Sales

Example:

Rank	Product	Sales
1	Laptop	₹450,000
2	Mobile	₹400,000
3	Table	₹126,000
4	Chair	₹111,000
5	Headphones	₹90,000

15. Salesperson Performance

Create a **Column Chart**.

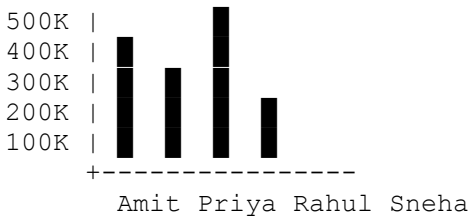
X-axis

Salesperson

Y-axis

Total Sales

Example:



This visualization identifies high-performing salespersons.

16. Sales and Profit by Category

Create a **Clustered Column Chart**.

X-axis

Category

Values

- Total Sales
- Total Profit

This allows students to compare:

Revenue vs. Profit

For example:

Category	Sales	Profit
Electronics	₹850K	₹105K
Furniture	₹200K	₹32K

Accessories ₹137K ₹20K

17. Add Slicers

Add the following slicers:

Slicer 1

Region

Slicer 2

Category

Slicer 3

Salesperson

Slicer 4

Order Date

Now the dashboard becomes interactive.

For example, selecting:

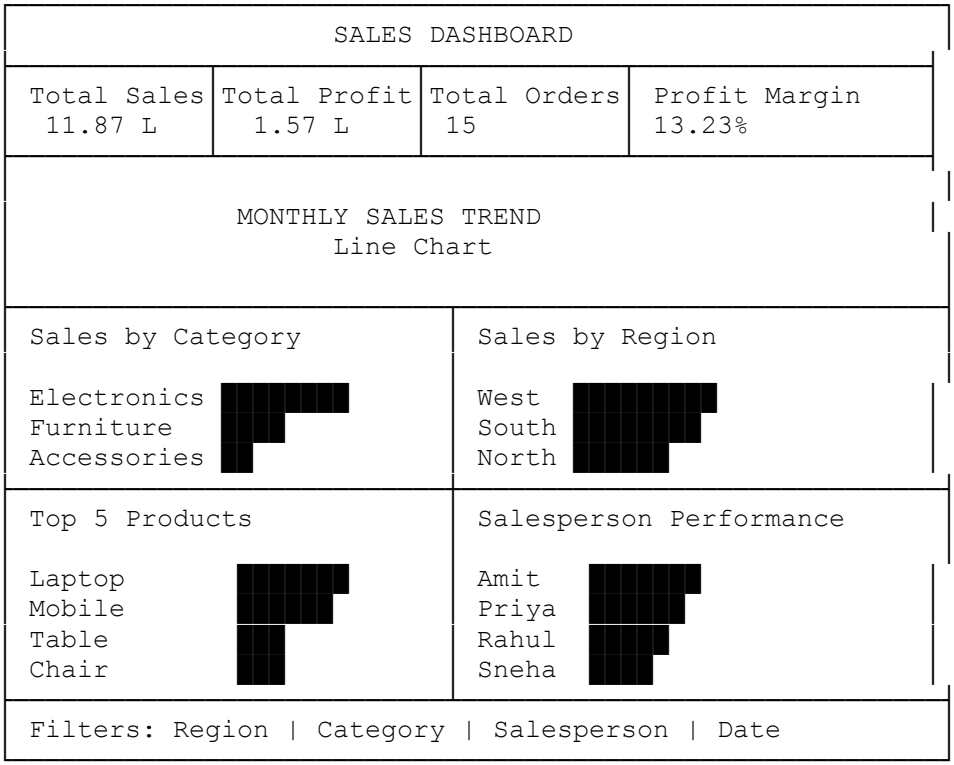
Region = West

will automatically update:

- Total Sales
- Total Profit
- Sales Trend
- Category Sales
- Product Sales
- Salesperson Performance

18. Dashboard Layout

The final dashboard can have the following arrangement:



19. Expected Analysis

After creating the dashboard, students should answer the following questions.

Q1. Which category has the highest sales?

Answer: Identify the category with the maximum sales from the Category chart.

Q2. Which region has the highest sales?

Answer: Identify the region with the highest sales from the Region chart.

Q3. Which product is the best-selling product?

Answer: Identify the product with the highest sales.

Q4. Which salesperson has generated the highest sales?

Answer: Identify the salesperson with the maximum sales.

Q5. Which month has the highest sales?

Answer: Identify the peak point in the Monthly Sales Trend.

Q6. What is the overall profit margin?

Answer:

```
Profit Margin =
Total Profit / Total Sales × 100
```

Q7. What happens to the dashboard when the West region is selected?

Students should observe how all visualizations change dynamically.

20. Example Business Insights

A student may write the following observations:

Business Insights

1. Electronics is the highest-performing product category.
2. The West region generates the highest sales.
3. Laptop is one of the highest-selling products.
4. Amit has generated the highest sales among the salespersons.
5. Sales show an increasing trend during the later months.
6. Electronics contributes significantly to overall profit.
7. Interactive filters allow management to analyze sales for a particular region, category, or salesperson.

21. Conclusion

The **Sales Dashboard** provides an interactive visual representation of sales performance. Power BI enables users to integrate data, clean data, perform calculations using DAX, create visualizations, and interactively analyze business performance.

The dashboard helps management make **data-driven decisions** regarding products, regions, salespersons, revenue, and profitability.

22. Student Submission

Students should submit:

1. **Excel dataset** – `Sales_Data.xlsx`
2. **Power BI file** – `Sales_Dashboard.pbix`
3. **Screenshots of the final dashboard**
4. **Short report containing business insights**
5. Answers to the seven analysis questions

Suggested Marks: 10

Component	Marks
Data import & cleaning	2
DAX measures	2
Appropriate visualizations	2
Interactive dashboard & slicers	2
Analysis and business insights	2
Total	**10**

This assignment is suitable for your **Data Science & Visualization practical**, particularly for demonstrating **data cleaning → analysis → visualization → interactive dashboard → interpretation** as one complete workflow.